

The background image shows a laptop screen with a data dashboard. At the top, a line graph shows two data series: 'New Visitor' (dark blue) and 'Returning Visitor' (light green). The 'New Visitor' line shows a significant peak around the 19th day of the month. Below the graph is a pie chart, mostly dark blue. The laptop's keyboard is visible at the bottom. The title 'MASS SHOOTINGS' is overlaid in large white text.

# MASS SHOOTINGS

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# Thesis Statement:

Age and variations in gun laws across the United States influence the severity of Mass Shootings.

# Project Objectives

- Analyze patterns in U.S. mass shootings to explore:
  - Changes over time
  - Influence of demographics (age, race, gender) and social factors (mental health) on shooter profiles
  - Possibility of predicting the type of firearm used
  - How fatalities compare among different states

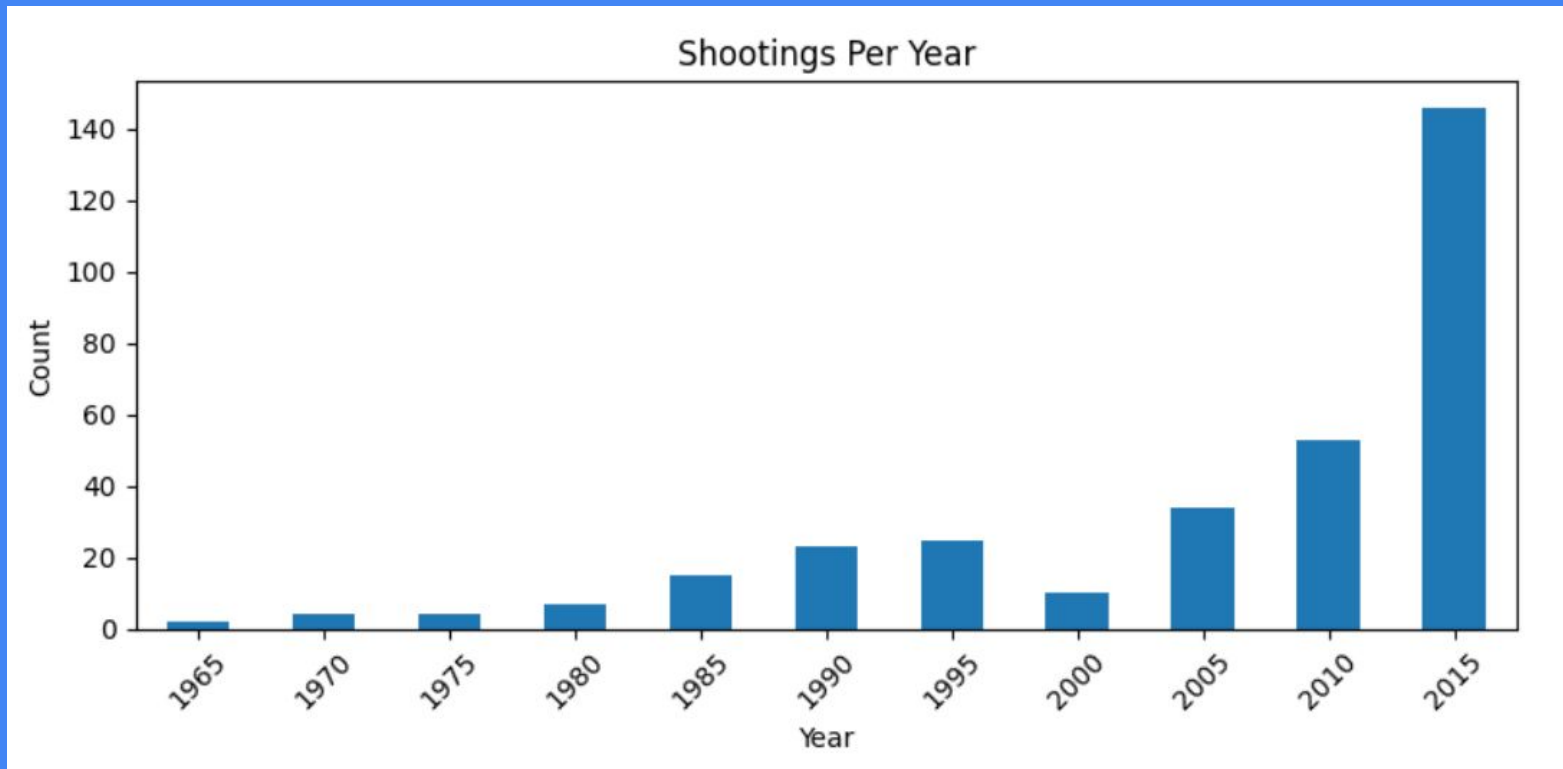
# Data Pre-Processing and Analytical Methods Used

- Data Cleansing
- Decision Tree Classifier
- Severity Ratios
- Heat Map Graph
- Cluster Analysis

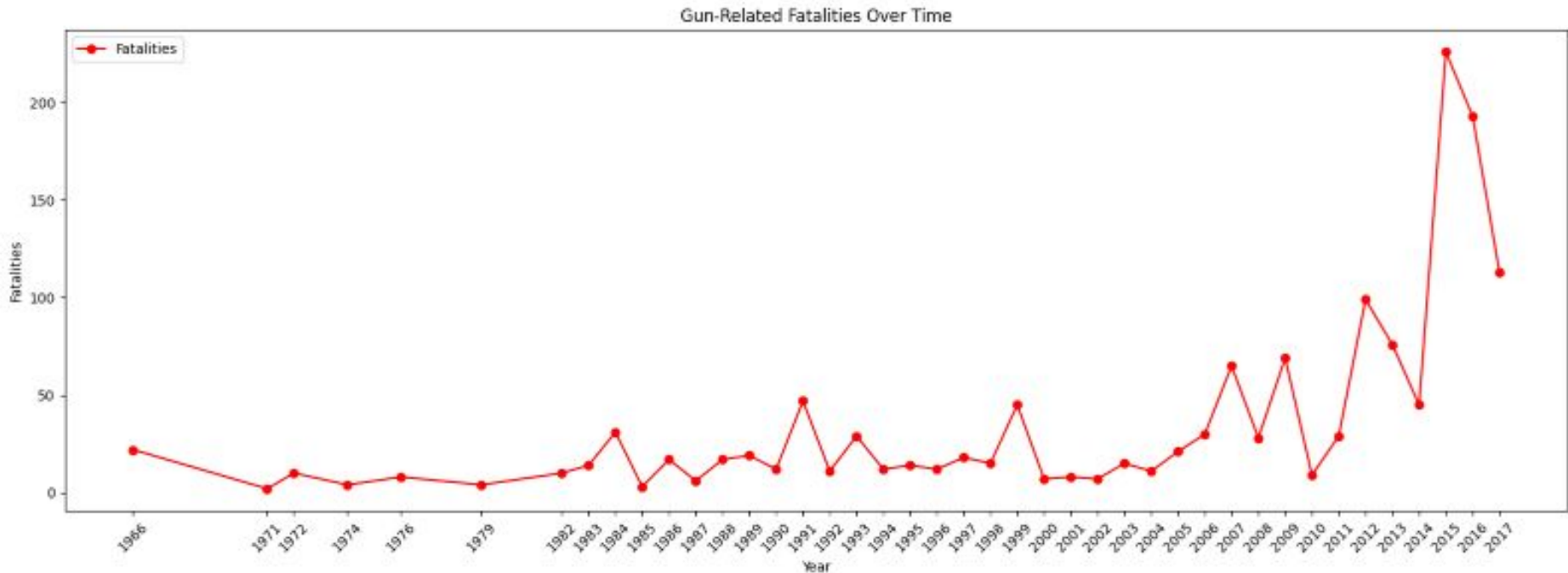
# Challenges

- Cleaning Weapon/Type Column
  - Gun Type Standardization
    - Issue: Inconsistent Values
      - (semi-automatic handgun, .38 revolver, semi-automatic rifle....)
    - Solution: Cleaned/standardized values
      - Assault Rifle, Handgun, Shotgun
- Low Number of Incidents Data present

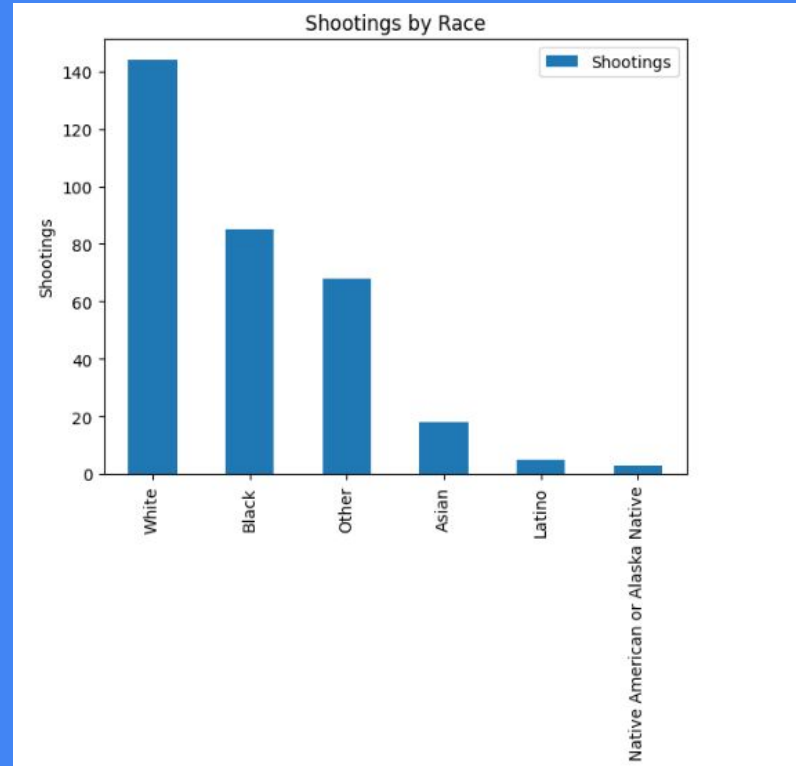
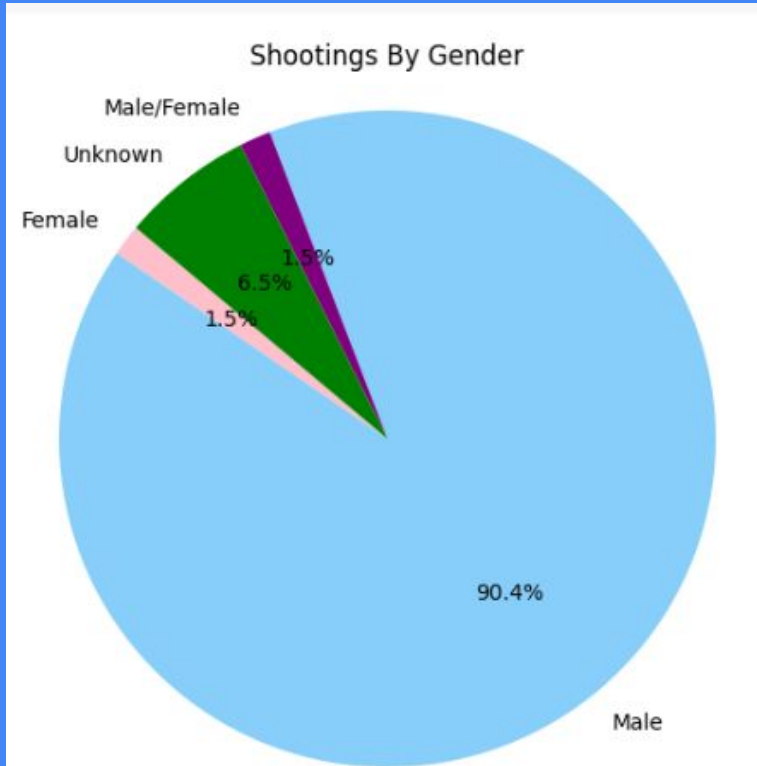
# Changes over time



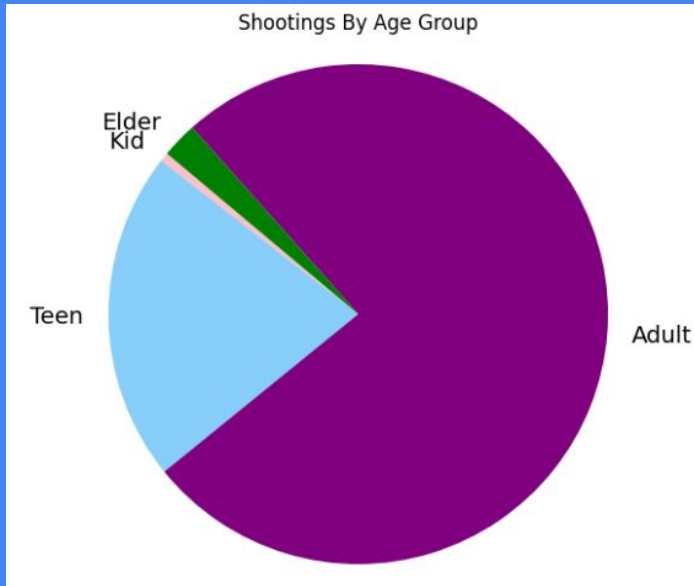
# Fatalities over time



# Gender, Race



# Age, Mental Health, Employment Status



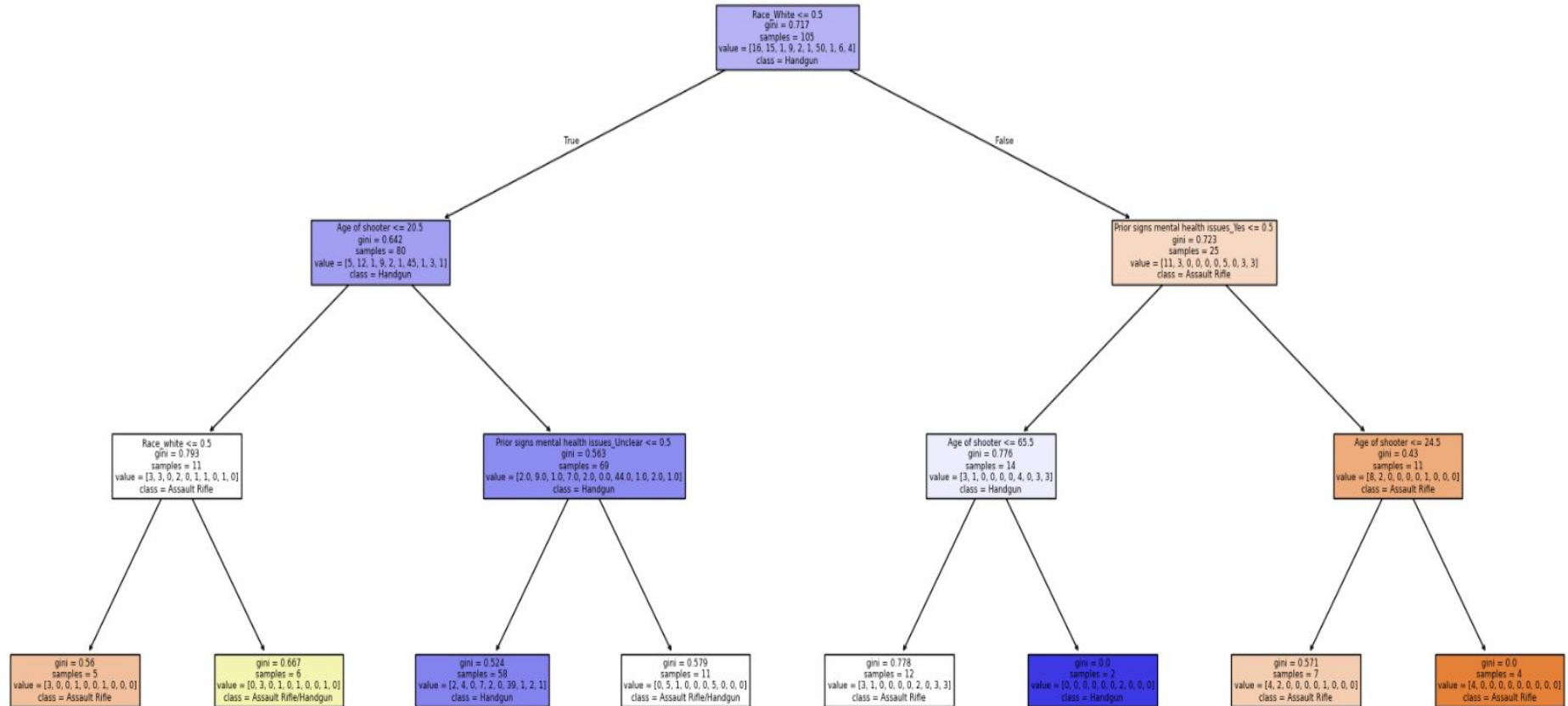
## Mental Health Issues

|         |     |
|---------|-----|
| Unclear | 124 |
| Yes     | 106 |
| No      | 93  |

## Employed (Y/N)

|     |    |
|-----|----|
| 1.0 | 42 |
| 0.0 | 25 |

# What Determines the Type of Weapon a Shooter will use?



# Metrics and Feature Importance

Accuracy: 0.41304347826086957

Confusion Matrix:

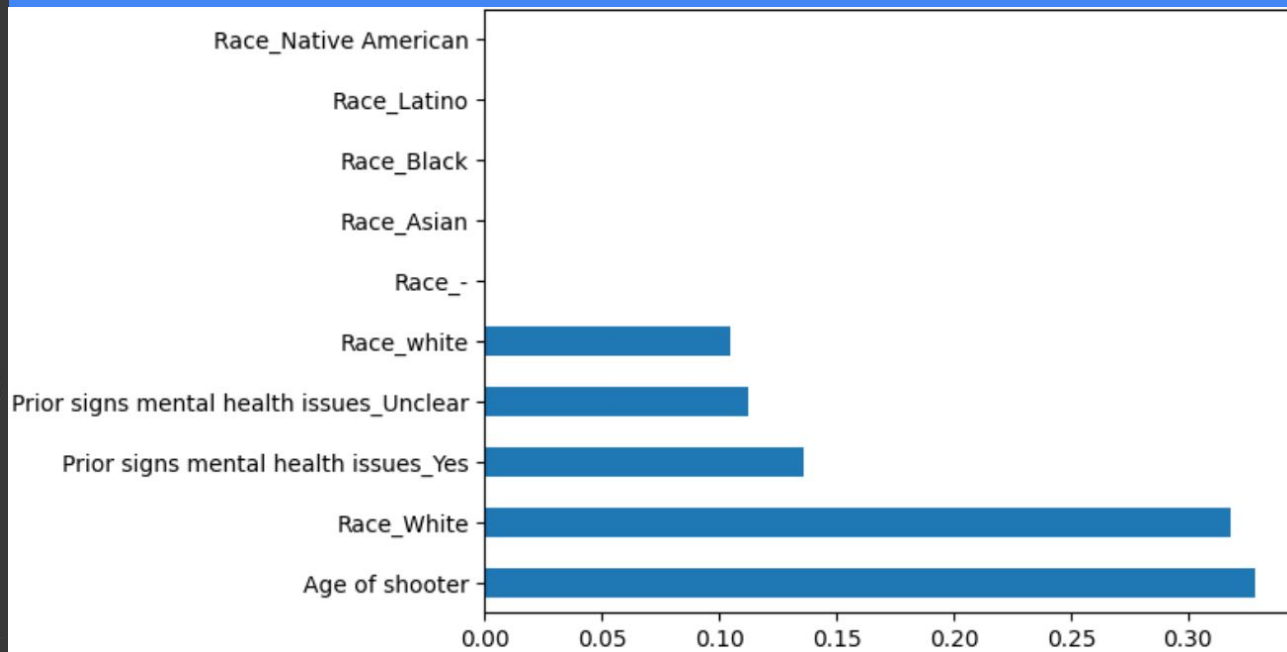
```
[[ 5  0  0  0  0  0]
 [ 5  3  0  6  0  0]
 [ 0  1  0  2  0  0]
 [ 5  2  0 11  0  0]
 [ 1  0  0  0  0  0]
 [ 3  0  0  2  0  0]]
```

Precision: 0.3857469761359922

Recall: 0.41304347826086957

f1: 0.35732998885172795

Error Rate 0.5869565217391304



# Fatality Ratios Across States

*We looked at how deadly mass shootings are in each state by finding the average number of fatalities per event. This helps us focus on severity, not just how often they happen.*

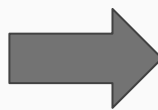
Why?

| fatalities  |     |
|-------------|-----|
| State       |     |
| California  | 178 |
| Texas       | 159 |
| Florida     | 129 |
| Nevada      | 68  |
| Colorado    | 53  |
| Virginia    | 53  |
| Connecticut | 41  |
| New York    | 40  |

MotherJones Dataset

| State       |     |
|-------------|-----|
| California  | 176 |
| Texas       | 143 |
| Florida     | 119 |
| Washington  | 78  |
| Nevada      | 72  |
| Virginia    | 48  |
| Colorado    | 46  |
| Arizona     | 44  |
| Connecticut | 42  |

Kaggle Dataset



Average fatalities per mass shooting per state

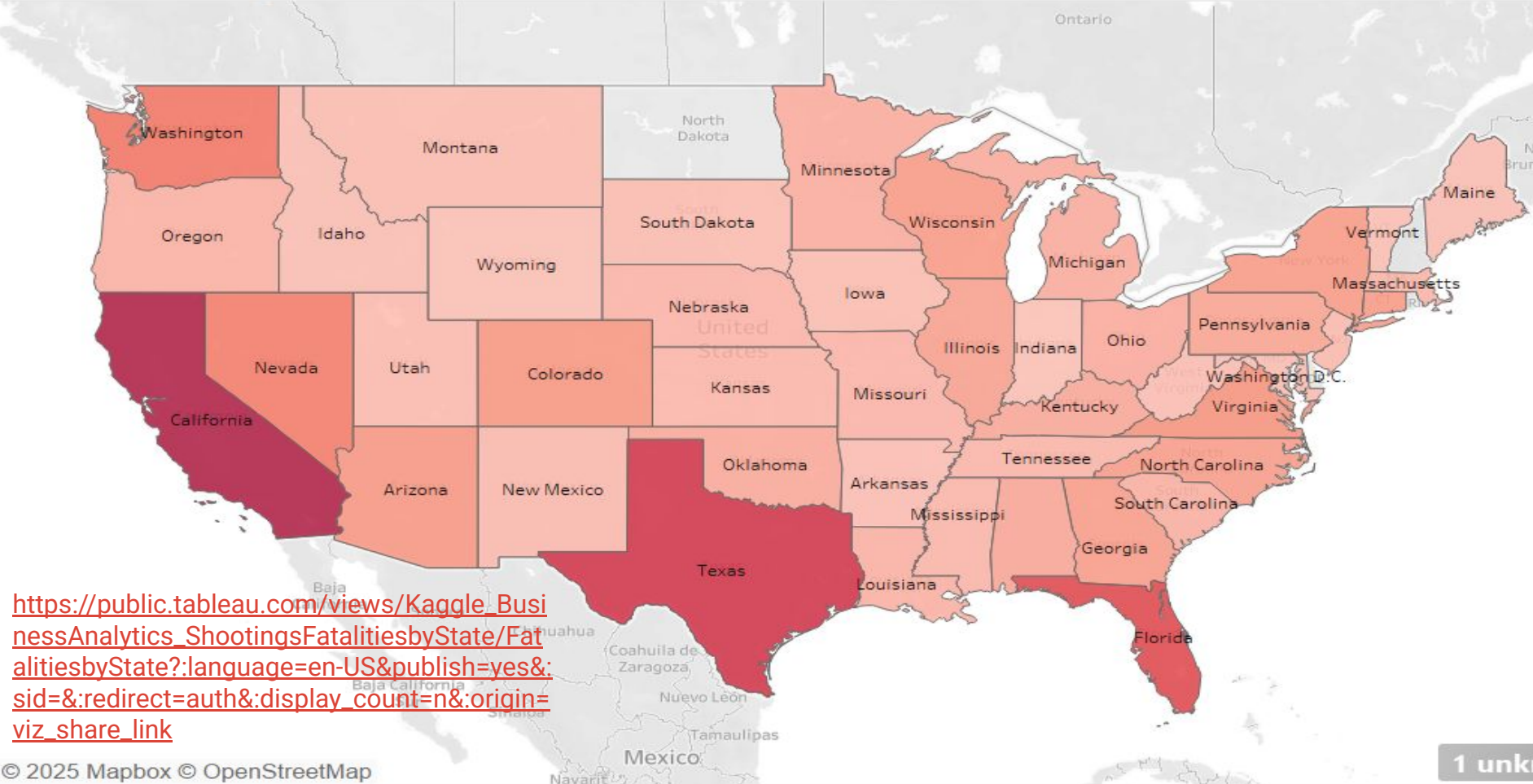
Nevada: 1:23  
Maine: 1:18  
Connecticut: 1:14  
Virginia: 1:13  
Texas: 1:12  
D.C.: 1:12  
Florida: 1:10  
Oklahoma: 1:10

MotherJones Dataset

Connecticut: 1:14  
Washington D.C.: 1:13  
Nevada: 1:12  
Virginia: 1:8  
Texas: 1:7  
Hawaii: 1:7  
Colorado: 1:6  
Iowa: 1:6  
South Dakota: 1:6  
Florida: 1:5  
California: 1:5

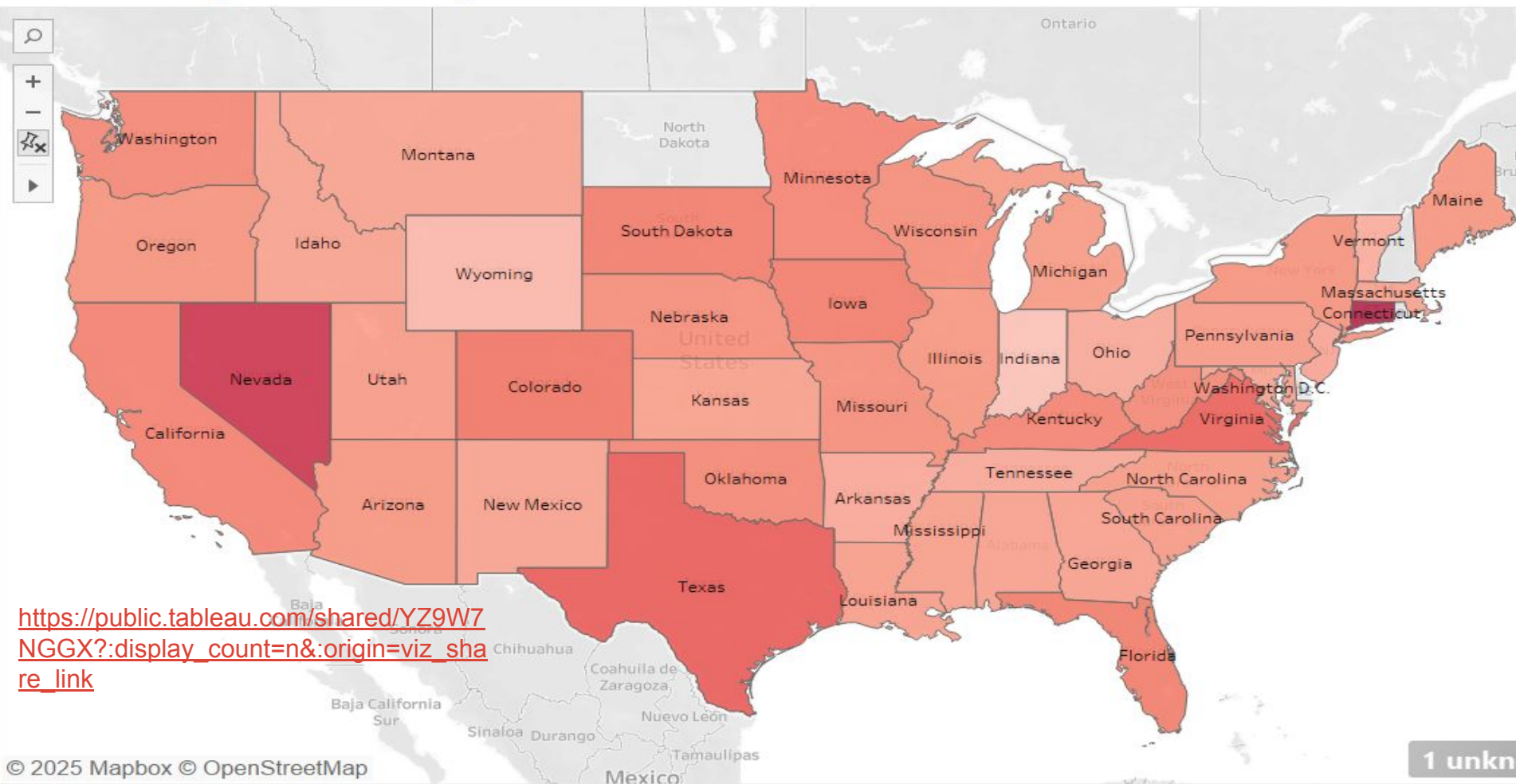
Kaggle Dataset

# Fatalities by State



[https://public.tableau.com/views/Kaggle\\_BusinessAnalytics\\_ShootingsFatalitiesbyState/FatalitiesbyState?:language=en-US&publish=yes&:sid=&:redirect=auth&:display\\_count=n&:origin=viz\\_share\\_link](https://public.tableau.com/views/Kaggle_BusinessAnalytics_ShootingsFatalitiesbyState/FatalitiesbyState?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link)

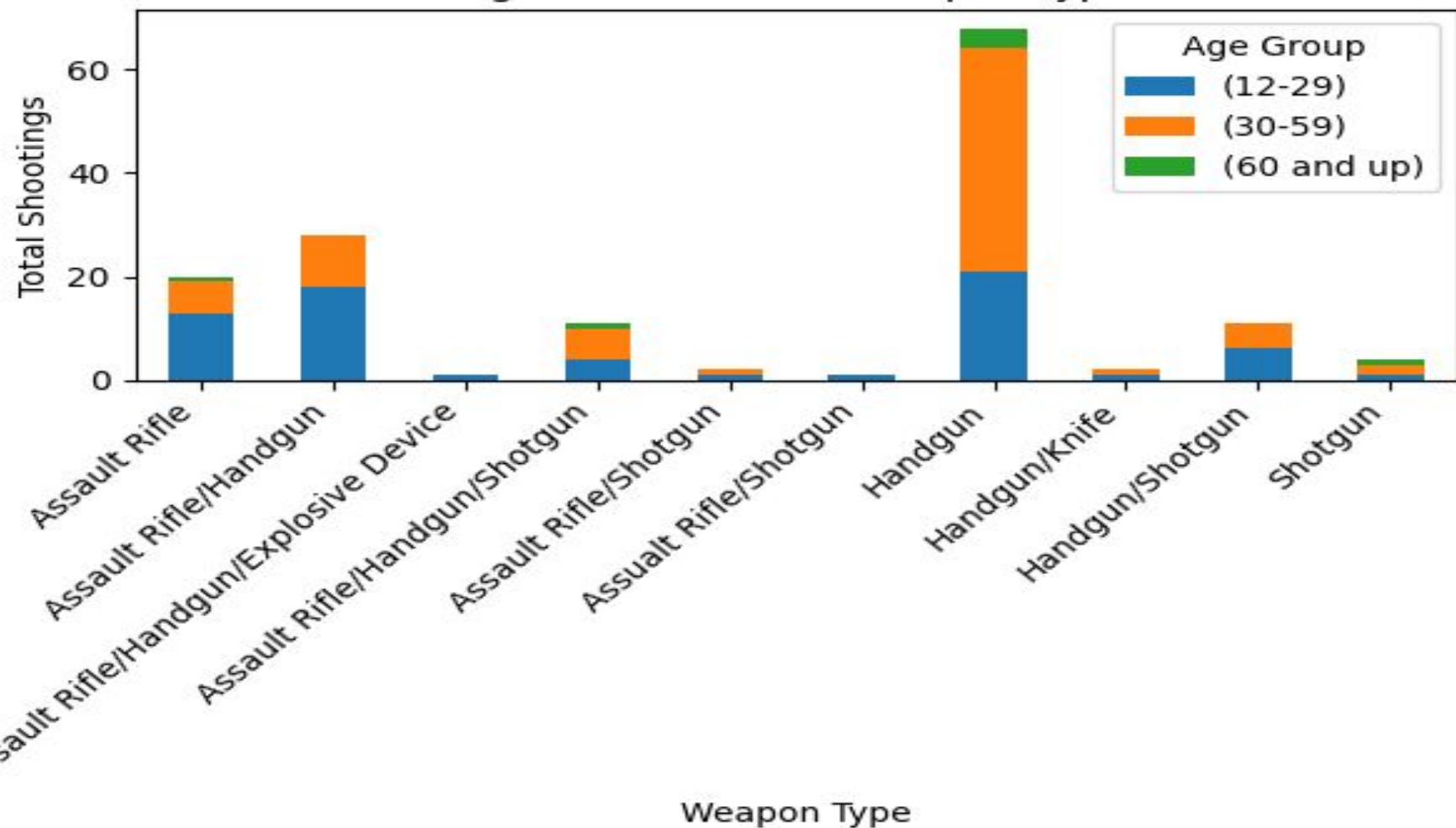
## Fatalities per Shooting



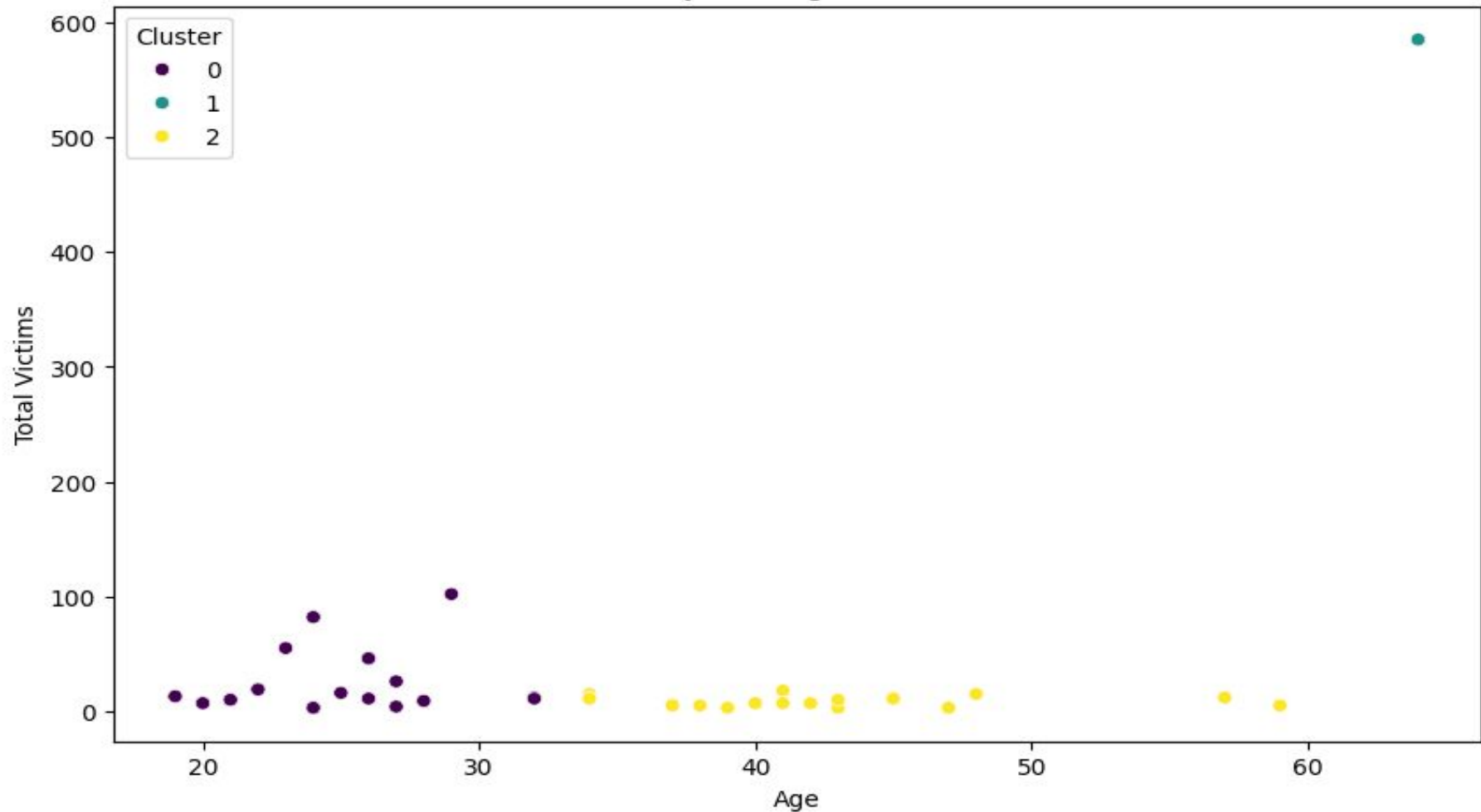
# Overall Significance

- *States like Nevada, Virginia, Maine have “relaxed” gun laws which in some cases allow you to open carry without a permit. This could lead to higher fatality ratios.*
- *Gun laws may affect severity, not just frequency.*
- *Note that some states may be skewed by one event, depending on magnitude.*

Age of Shooter vs. Weapon Type



Cluster Analysis of Age vs. Total Victims



## Recommendations:

- Implement Stricter Gun Laws in High-Risk States
- Age-Based Firearm Restrictions
- Expand Mental Health Screenings and Early Intervention
- Enhance Security Protocols in Schools and Youth Spaces